[Khanh Pham] Boot Process (ARM), U-Boot, Linux Kernel, Root Filesystem & Buildroot

Boot process in ARM

As we know, OS is component that manage and control both hardware and application of every device. However, before running OS, there are many step from power on the device to the OS load to RAM. In this document,I will focus on the boot process of ARM chip architecture

Overview boot process

Before writing about the boot process, we must understand about some important definitions:

- **BootRom** is a small piece which is writen by chip-vendor inside the processor chip. It contains the very first code which is executed by the processor on power-on or reset. It decide where to load the next part of code to be executed.
- **Bootloader** is the component which is responsible for finding and loading the final OS or firmware which is supposed to run on the chip

Now we will go to the overview of boot process in ARM architecture.

Bootrom stage

First of all, after we power on the device, CPU will load the fixed address where the bootRom inside. The purpose of bootrom stage is:

- BootRom performs the essential initialization including programming the clocks, stacks
- Checking the boot strapping pin to decide boot device: SD card, eMMC, NAND Flash or USB

In this moment, the main RAM(DRAM) isn't start, so bootrom can't load all of bootloader. Because it can only use the SRAM, so we must go to next step:

Second program loader stage

In this stage, because bootrom can't load all of bootloader, it only loads the minimal of loader, we call it SPL.

The main purpose of the second stage is: **Initializing the main RAM(DRAM)**

After the DRAM is initialized, we can load the main bootloader to RAM. With arm architecture, we call it `Uboot`.

Then we can go to the last stage of boot process: `Uboot stage`

Uboot stage

After SPL completed and change to jump to Uboot

This stage have some purposes:

- Initializing all other devices like: network, usb, screen,...
- Load the environment variable and CLI of Uboot
- Load the OS component: Kernel, device tree blob

After all, kernel will take the control from uboot and linux kernel start. Then the first process(init) will be start and device start completed

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